

India FMCG Sales & Profitability Analytics Dashboard

Power BI Project Report | FY 2023–24

1. Executive Summary

This project presents an end-to-end Power BI analytics solution developed to analyze FMCG sales performance across India during FY 2023–24. The dashboard integrates revenue, profitability, customer, product, and regional performance metrics into a single interactive reporting platform.

The analysis covers ₹94 million in sales, ₹16 million in profit, over 127,000 units sold, and operations across 30+ Indian states. The dashboard enables users to track quarterly performance, identify profit-driving segments, and evaluate the impact of discounting on profitability.

Three key business findings emerged:

- Enterprise customers generated negative profit margins (-2% to -5%).
 - Discounts above 5% consistently reduced profitability.
 - Madhya Pradesh was the largest contributor to overall profit.
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2. Business Problem

The organization lacked a centralized analytics solution for monitoring profitability across products, customer segments, and geographies.

Key challenges included:

- Limited visibility into state-wise profit contribution.
 - Manual Excel-based quarterly reporting.
 - No structured analysis of discount impact on margins.
 - Difficulty identifying profitable and underperforming customer segments.
 - Lack of a consolidated executive reporting view.
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3. Project Objectives

The primary objectives were:

1. Build an interactive Power BI dashboard for FY 2023–24 sales analysis.
 2. Enable state-level, product-level, and segment-level profitability analysis.
 3. Quantify the relationship between discounts and profit margins.
 4. Provide quarterly performance tracking through Q1–Q4.
 5. Deliver executive-ready KPI reporting for business stakeholders.
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4. Dataset Overview

Metric	Value
Period Covered	FY 2023–24
Total Revenue	₹94M
Total Profit	₹16M
Profit Margin	17%
Units Sold	127K+
Transactions	77
States Covered	30+
Customer Segments	5
Product Categories	5

Key Dimensions

- State
- Customer Segment
- Product Category
- Quarter
- Year

Key Measures

- Net Sales
 - Total Profit
 - Profit Margin %
 - Units Sold
 - Discount %
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5. Data Preparation

The dataset was imported from Excel and transformed using Power Query.

Data Cleaning Activities

- Removed duplicate transaction records.
- Validated missing values across critical fields.
- Standardized state and segment names.
- Enforced consistent data types.
- Derived Quarter and Year attributes from transaction dates.

Data Modeling

A star schema architecture was implemented consisting of:

- Sales Fact Table
- State Dimension
- Segment Dimension
- Product Dimension
- Date Dimension

This design improved query performance and enabled efficient cross-filtering across visuals.

6. Dashboard Development

Tools Used

Tool	Purpose
Power BI Desktop	Dashboard Development
Power Query	Data Transformation
DAX	KPI Calculations
Excel	Data Source
Data Modeling	Star Schema Design

Key DAX Measures

- Total Net Sales
- Total Profit
- Profit Margin %
- Total Units Sold

- Total Transactions
- State Profit Rank
- Quarter-to-Date Sales
- Average Discount Rate

A total of 15+ DAX measures were developed to support dynamic reporting and KPI calculations.

7. Dashboard Features

KPI Summary

The dashboard provides five dynamic KPI cards:

- Total Revenue
- Total Profit
- Profit Margin %
- Units Sold
- Transaction Count

Geographic Performance Analysis

A treemap visual highlights state-level profit contribution and identifies high-performing regions.

Customer Segment Analysis

A horizontal bar chart compares profitability across customer segments and identifies margin leaders and laggards.

Product Performance Analysis

Product category margins are tracked to identify seasonal and category-level performance trends.

Discount Impact Analysis

A scatter plot visualizes the relationship between discount percentage and profit margin.

Quarterly Performance Tracking

Interactive quarter filters allow users to compare Q1–Q4 performance and identify trends over time.

8. Key Findings

8.1 Revenue and Profit Overview

- Total Revenue: ₹94M
 - Total Profit: ₹16M
 - Overall Profit Margin: 17%
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8.2 Geographic Insights

- Madhya Pradesh generated the highest profit contribution.
 - Telangana and Jharkhand ranked among the top-performing states.
 - States such as Goa and Mizoram achieved exceptionally high margins despite lower sales volumes.
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8.3 Customer Segment Insights

Segment	Observation
Channel Partners	Highest margin (~75%)
Midmarket	Consistent profitability
Government	Moderate performance
Small Business	Low but positive margin
Enterprise	Negative margin (-2% to -5%)

Enterprise was the only segment that consistently destroyed value.

8.4 Product Insights

- Textiles led profitability in Q1 and Q4.
 - Steel Products performed strongest in Q2.
 - Basmati Rice peaked during Q3.
 - Product profitability demonstrated clear seasonal variation.
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8.5 Discount Analysis

The analysis revealed a strong negative relationship between discounts and profit margins.

Key observation:

Transactions with discounts greater than 5% consistently produced lower margins compared to transactions with lower discount rates.

8.6 Quarterly Performance

Quarter	Revenue	Margin
Q1	₹16M	22%
Q2	₹30M	19%
Q3	₹21M	17%
Q4	₹27M	13%

Q1 delivered the highest profitability, while Q4 generated the highest revenue but the lowest margin.

9. Business Recommendations

Recommendation 1: Expand Channel Partner Business

Channel Partners consistently generated the strongest margins and should be prioritized for future growth.

Recommendation 2: Review Enterprise Pricing Strategy

The Enterprise segment requires immediate pricing and contract review due to persistent negative profitability.

Recommendation 3: Establish Discount Controls

Implement a discount approval process and maintain discounts below 5% whenever possible.

Recommendation 4: Investigate Q4 Margin Erosion

Conduct deeper analysis to determine whether pricing concessions or product mix changes drove declining margins.

Recommendation 5: Focus on High-Margin States

Increase investment in efficient markets such as Goa, Mizoram, and Daman & Diu.

10. Challenges Encountered

- Managing DAX filter context across multiple slicers.
 - Validating negative-margin Enterprise transactions.
 - Standardizing state-level geographic data.
 - Maintaining scatter plot readability with overlapping observations.
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11. Skills Demonstrated

Business Intelligence

- KPI Development
- Performance Analytics
- Executive Reporting

Power BI

- Interactive Dashboard Design
- Advanced Visualizations
- Drill-Down Analysis

Data Modeling

- Star Schema Design
- Fact-Dimension Modeling

DAX

- KPI Measures
- Ranking Logic
- Time Intelligence Functions

Data Preparation

- Power Query Transformations
 - Data Cleaning
 - Data Validation
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12. Conclusion

This project demonstrates the application of Power BI, DAX, Power Query, and dimensional modeling to transform transactional sales data into actionable business intelligence.

The dashboard identified a loss-making customer segment, quantified the impact of discounting on profitability, and highlighted key geographic profit drivers. By consolidating revenue and profitability reporting into a single platform, the solution enables faster decision-making and supports data-driven commercial strategy across the FMCG business.